JEE APEX FTS – JEE ADV. PART TEST-02 [P1]

KEY SHEET

MATHEMATICS

1	**B**	2	**D**	3	**A**	4	**C**	5	**ACD**	6	**ABCD**
7	**BC**	8	**52**	9	**4**	10	**6**	11	**80**	12	**2015**
13	**21**	14	**C**	15	**B**	16	**D**	17	**C**		

PHYSICS

18	**A**	19	**A**	20	**A**	21	**B**	22	**BC**	23	**BC**
24	**CD**	25	**4**	26	**2**	27	**2**	28	**3**	29	**3**
30	**5**	31	**D**	32	**A**	33	**A**	34	**B**		

CHEMISTRY

35	**B**	36	**B**	37	**C**	38	**D**	39	**BD**	40	**ABC**
41	**ABCD**	42	**4**	43	**396**	44	**6**	45	**65**	46	**7**
47	**4**	48	**B**	49	**A**	50	**D**	51	**A**		

SOLUTIONS
MATHEMATICS

1. ${}^5C_1$ – ANY ONE OF THE FIVE PERSONS

${}^4C_2$ - ANY TWO OF THE FRUIT VARIETY [EXAMPLE APPLE, BANANA]

2!-ONE OF THEM MAY GET APPLES OR BANANAS

P3	P4	P5
1APPLE	1BANANA	2 ORANGES
3 CHICKOOS	3 ORANGES	2 CHICKOOS

3!-THREE SET CAN BE GIVEN TO THREE PERSONS AND

2!-CHICKOOS AND ORANGES CAN BE INTERCHANGED

TOTAL ${}^5C_1.{}^4C_2.2!3!2! = 1440$

2. HERE MEAN IS 15/4

$$9.\ t = -2 - x$$

$$\int_{-2-a}^{-2-b} \ln\left(\frac{1+t+t^2}{3+3t+t^2}\right)dt = 0$$

$\Rightarrow\ a = -2 - a \qquad -2 - b = b$

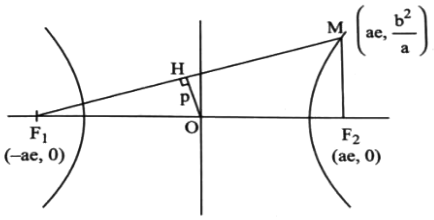
$a = -1 \qquad b = -1$

$|a+b| = 2.$

3. $P(X \cap Y) = K \Rightarrow P(Y) = 2K, P(X) = 3K \Rightarrow P(X\Delta Y) = 3k\ \&\ k \neq 0$

$P(X \cup Y) = 4k \leq 1 \qquad$ If $k = 6k^2 \Rightarrow k = \dfrac{1}{6}$

4.



$\dfrac{OH}{OF_2} = \lambda \in \left(\dfrac{1}{3}, \dfrac{1}{2}\right) e \in ?$ $\qquad$ Equation of $HF_1 : y = \dfrac{b^2}{a(2ac)}(x + ac)$

$y2a^2e = b^2x + aeb^2$ $\qquad$ $b^2x - 2a^2ey + aeb^2 = 0$

$OH = p = \dfrac{aeb^2}{\sqrt{b^4 + 4a^4e^2}}$ $\qquad$ $OF_2 = ae$

$$\frac{OH}{OF_2} = \lambda = \frac{b^2}{\sqrt{b^4 + 4a^4e^2}} = \frac{b^2}{\sqrt{b^4 + \dfrac{4b^4}{(e^2-1)^2}e^2}} = \frac{1}{\sqrt{1 + \dfrac{4e^2}{(e^2-1)^2}}}$$

$\lambda = \dfrac{e^2 - 1}{e^2 + 1} \qquad \underbrace{\dfrac{1}{3} < \dfrac{e^2 - 1}{e^2 + 1}} < \dfrac{1}{2}$

LHS: $e^2+1<3e^2-3$

RHS: $24<2e^2$ $\quad e^2<3$ $\quad \sqrt{2}<e<\sqrt{3}$

5. $f(x)=\int_0^x\left(5\ell n\left(1+t^2\right)-10t\tan^{-1}t+16\sin t\right)dt$

$f'(x)=5\ell n\left(1+x^2\right)-10x\tan^{-1}x+16\sin x \qquad f'(0)=0$

$f''(x)=\frac{10x}{1+x^2}-\frac{10x}{1+x^2}-10\tan^{-1}x+16\cos x$

$=16\cos x-10\tan^{-1}x \ >0,\ \forall x\in(0,1)$, **implies** $f'(x)$ **is increasing**

$f'(x)=0$ for some $x\in(0,1)$

But $f'(0)=0, f'(1)>0$ So from $(0,1)\ f'(x)>0$

$f(x)$ is $\uparrow$ function.

6. Let z_1,z_2 be the roots, $|z_1|=1\ z_2=\frac{c}{a}\frac{1}{z_1}\Rightarrow|z_2|=\frac{|c|}{|a|}$

$z_1+z_2\ \frac{-b}{2}\ \&\ |a|=|b|\Rightarrow|z_1+z_2|^2=1 \qquad (z_1+z_2)\left(\overline{z_1}+\overline{z_2}\right)=1\Rightarrow(z_1+z_2)\left(\frac{1}{z_1}+\frac{1}{z_2}\right)=1$

$\Rightarrow(z_1+z_2)^2=z_1\,z_2\left(\frac{-b}{a}\right)^2=\frac{c}{a} \qquad \Rightarrow b^2=ca$

$\Rightarrow(a-b)^2+(b-c)^2+(c-a)^2=0$

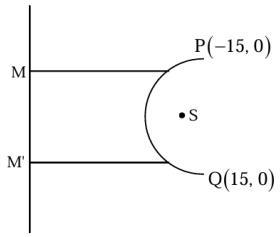
$(a-b)^2+(b-c)^2+(c-a)^2+2(a-b)+(b-c)^2=2(a-b)(b-c)$

$\Rightarrow(a-b)^2+(a-b)(b-c)$

& Let $\alpha=|b-c|,\ \beta=|c-a|,\ \gamma=|a-b|$

$\Rightarrow\beta^2=\gamma\alpha,$ parallel $\gamma^2=\alpha\beta,\alpha^2=\beta\gamma \qquad \Rightarrow(\alpha-\beta)^2+(\beta-\gamma)^2+(\gamma-\alpha)^2=0\ \alpha=\beta=\gamma$

7. Chord passes through $P(-15,0)\ and\ Q(15,0)$



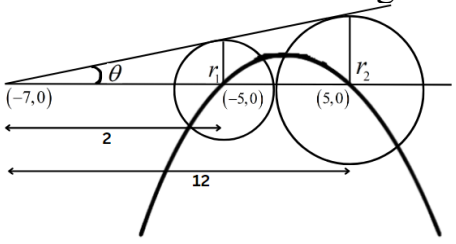
Let S be the focus $SP=PM$

Takes PM as radius and P as centre draw a circle similarly QM' as radius and Q as centre draw a circle.

If circles are intersecting, two foci are possible $\Rightarrow$ two parabolas exist

$PM+PM'>PQ$

If two circles are touching each other, then only one focus is possible



$\frac{r_1}{r_2}=\frac{1}{6}$ $\quad r_1+r_2=10 \quad r_1=\frac{10}{7}, r_2=\frac{60}{7}$ $\quad \sin\theta=\frac{5}{7}$

$\tan\theta=\frac{5}{2\sqrt{6}}$ $\quad x+my+7=0 \quad \frac{-1}{m}=\frac{5}{2\sqrt{6}}$ $\quad m=\frac{-2\sqrt{6}}{5}$

Hence $m\in\left(\frac{-2\sqrt{6}}{5},\frac{2\sqrt{6}}{5}\right)-\{0\}$

8. $703-x-y=xy \quad \Rightarrow (x+1)(y+1)=704\left(=2^6\times 11\right)$

$=22\times 32 \quad \Rightarrow x=21, y=31$

9. $\int_{-1}^{0}1007dx+\int_{0}^{1}1008dx=2015$

10. $52\xrightarrow{\text{face card removed}}40\xrightarrow{\text{20 drawn randomly}}$

LET E_o : 20 CARDS RANDOMLY REMOVED HAS 4 ACES.

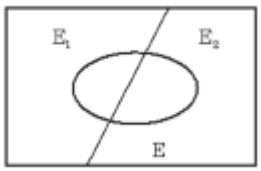
LET E_1 : 20 CARDS RANDOMLY REMOVED HAS 3 ACES.

LET E_2 : 20 CARDS RANDOMLY REMOVED HAS 2 ACES.

E : EVENT THAT 2 DRAWN FROM THE REMAINING 20 CARDS HAS BOTH THE ACES.

$P(E)=P(E\cap E_o)+P(E\cap E_1)+P(E\cap E_2)$

$=P(E_o).P\left(\frac{E}{E_o}\right)+P(E_1).P\left(\frac{E}{E_1}\right)+P(E_2).P\left(\frac{E}{E_2}\right)=40\begin{bmatrix}4\text{aces}\\36\text{other}\end{bmatrix}$



$=\frac{{}^4C_0.{}^{36}C_{20}}{{}^{40}C_{20}}.\frac{{}^4C_2}{{}^{20}C_2}+\frac{{}^4C_1.{}^{36}C_{19}}{{}^{40}C_{20}}.\frac{{}^3C_2}{{}^{40}C_{20}}+\frac{{}^4C_2.{}^{36}C_{18}}{{}^{40}C_{20}}.\frac{{}^2C_2}{{}^{20}C_2}$

$=\frac{\left({}^{36}C_{20}.{}^4C_2\right)+\left({}^4C_1.{}^{36}C_{19}.{}^3C_2\right)+\left({}^4C_2.{}^{36}C_{18}.{}^2C_2\right)}{{}^{40}C_{20}.{}^{20}C_2}$

$=\frac{6.{}^{36}C_{20}+12.{}^{36}C_{19}+6.{}^{36}C_{18}}{{}^{40}C_{20}.{}^{20}C_2} \quad =\frac{6\left[{}^{36}C_{20}+2{}^{36}C_{19}+{}^{36}C_{18}\right]}{{}^{40}C_{20}.{}^{20}C_2}$

$=\frac{6\left({}^{37}C_{20}+{}^{37}C_{19}\right)}{{}^{40}C_{20}.{}^{20}C_2}=\frac{6\left({}^{38}C_{20}\right)}{{}^{40}C_{20}.{}^{20}C_2}\Rightarrow p=6$

11. Let $b-a\geq 6$

$1\leq acb<10\Rightarrow 1\leq a<b-5<5 \quad \Rightarrow {}^5C_2=10$

Similarly for $a-b\geq 6$, we get 10 $\quad \therefore 100-10-10=80$

12. $\int_{-1}^{0}1007dx+\int_{0}^{1}1008dx=2015$

13. $\alpha+\beta=0, \alpha=0$ and $\alpha=4\beta$ are 3 possibilities but $\alpha+\beta=0, \alpha=0$ not possible

(∵ Vol. of parallelepiped > 0) $\alpha = 4\beta$ $\beta = \begin{vmatrix} 1 & 0 & -1 \\ 3c & a & -a \\ 0 & b & c \end{vmatrix}$; $\alpha = 4.\begin{vmatrix} 1 & 1 & 1 \\ c & a & b \\ b & c & a \end{vmatrix}$

Here $\beta = \pm(ac + ab - 3bc)$ but for $-ve$ case, we get no triplets

$\Rightarrow (b + c - a)^2 = 0$ [on simplifying] $\Rightarrow a = b + c$ and $a, b, c \in \{0,1,2,3,5,6,7,8,9\}$

$\Rightarrow$ Total 21 possible

14. $f(x) = (x-1)(x+2)\left(x^2 + (2a-1)x + 1\right) = 0 \& (2a-1)^2 - 4 < 0$

$\Rightarrow -\frac{1}{2}a < \frac{3}{2} \Rightarrow a = 1 \& P = \frac{1}{6}.$

15. $\frac{1}{y^3}\frac{dy}{dx} - \frac{1}{y^2}x = -(1+\cos x) \& u = -\frac{1}{y^2}$

$\frac{dy}{dx} + \frac{2}{x}u = 2(1+\cos x)$ $\qquad 4x^2 = -2\int (1+\cos x)dx$

$\frac{-x^2}{y^2} = \frac{x^3}{3} + x^2 \sin x + 2x\cos x = -2\sin x + K$

$\frac{-\pi^2 2}{\pi} = \frac{\pi^3}{3} + (-2\pi) + K \Rightarrow K = \frac{-\pi^3}{3}.$

16. $I_{1,10} = \frac{\sin 2x}{2}\left(1 - \tan^2 x\right)^{10} - \int 10\left(1 - \tan^2 x\right)^9 \left(-2\tan x \sec^2 x\right)\frac{\sin 2x}{2}dx$

$\Rightarrow$ 4. $I_{0,10} + \frac{I_{1,10}}{5} = \frac{\sin 2x}{10}\left(1 - \tan^2 x\right)^{10} + 4T_{0,9} + C$

17. A) R.N. of terms $= 26 + 24 + ... + 2 = 182$

B) R.N. of terms $= 25 + 23 + ... + 1 = 169$

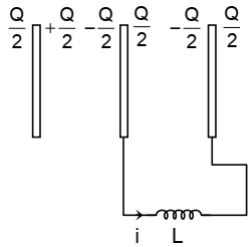
C) R.N. of terms $= 51 + 49 + ... + 1 = 676$

D) reg. coefficient $= f(1) \Rightarrow 4^{25} + 2^{25} = 2^{25}\left(2^{25} + 1\right)$

PHYSICS

18.

19. $\dfrac{\left(\dfrac{Q}{2}\right)^2}{2C}=\dfrac{1}{2}L_0^2 \Rightarrow i_0=\dfrac{Q}{2\sqrt{LC}}$



20. $\varepsilon=\pi r^2\left(\dfrac{dB}{dt}\right),\ dR=\dfrac{2\pi r}{\sigma\delta dr},\ dP=\dfrac{\varepsilon^2}{dR}=\dfrac{\pi\delta\sigma}{2}\left(\dfrac{dB}{dt}\right)^2 r^3dr,\ P=\dfrac{\pi\delta\sigma a^2}{8}\left(\dfrac{dB}{dt}\right)^2$

21. $\dfrac{1}{\lambda}=K\left(\dfrac{1}{4}-\dfrac{1}{n^2}\right)\Rightarrow-\dfrac{1}{\lambda^2}\dfrac{d\lambda}{dn}=+\dfrac{2k}{n^3}\Rightarrow\therefore\dfrac{d\lambda}{\lambda}=\dfrac{-2k\lambda}{n^3}$

$$\left|\frac{d\lambda}{\lambda}\right|=\frac{2k}{n^3\left(\frac{1}{\lambda}\right)}=\frac{2k}{n^3\left(k\left(\frac{1}{4}-\frac{1}{n^2}\right)\right)}=\frac{2}{\frac{n^3}{4}-n}\quad\Rightarrow\frac{\lambda}{d\lambda}=R=\left(\frac{\frac{n^3}{4}-n}{2}\right)\Rightarrow\frac{n^3}{4}-n<2R$$

22. Let I_1 & I_2 be current through B_1 & B_2 $I_1\times\sqrt{R^2+X_C^2}=220$

$\dfrac{I_1}{I_2}=\dfrac{\sqrt{R^2+X_C^2}}{\sqrt{R^2+X_C^2}}=\dfrac{\sqrt{R^2+(1/\omega C)^2}}{\sqrt{R^2+(\omega L)^2}}=\sqrt{\dfrac{R^2+40}{R^2+9.87}}$ $I_2>I_1$ B_2 will be brighter than B_1. As frequency increases, X_C decreases, X_L increases. So, I_2 becomes less than I_1. Hence brightness of B_1 will increases and that of B_2 will decreases

23. $\phi_{ABD}-\phi_{ACD}=-2\pi\ L\left[\dfrac{1}{\lambda_{AB}}-\dfrac{1}{\lambda_{CD}}\right]$ $\lambda=\dfrac{h}{p}=\dfrac{h}{\sqrt{2mE}}$ $E_{AB}>E_{CD}\Rightarrow\lambda_{AB}<\lambda_{CD}$

$\Rightarrow\phi_{ABD}-\phi_{ACD}<0$; matter wave following path ABD lags behind by other one following path ACD. For interference minima at D.

$\phi_{ACD}-\phi_{ABD}=(2n+1)\pi, n\in 0,1,2;$ for $n=1\ \phi_{ACD}-\phi_{ABD}=(2n+1)\pi, n\in 0,1,2;$ for $n=1$

$$\frac{2\pi\ L}{h}\left[\sqrt{2mE_0}-\sqrt{2\ m\left\{E_0-mgLsin\left(30^\circ\right)E_0-mgLsin\left(30^\circ\right)\right\}}\right]=\pi$$

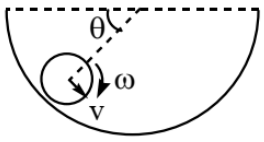
$\sqrt{2mE_0}-\sqrt{m(2E_0-mgL)}-\dfrac{h}{2\ L}=0;$ for first order minima

24. $\Delta V=\varepsilon-ir$ as the cell is discharging, So $\Delta V=($ Electrostatic field $)(0.1)=0.8$ V

$\varepsilon=\left(\dfrac{10\ N}{1C}\right)(0.1)=1$ V So $ir=0.2$ V $\Rightarrow r=0.2\Omega$

25. $\Delta x=(1.55-1.5)t=1000$ nm $\delta=\pi$, Intensity $=I_0+\dfrac{I_0}{4}+2\sqrt{I_0\left(\dfrac{I_0}{4}\right)}\cos\pi=\dfrac{I_0}{4}=\dfrac{I}{16}$

26. Magnetic forces due to angular motion of sphere will not create any torque. Magnetic forces due to translational motion passes through center of mass of sphere acts parallel to Normal reaction.



$v = r\omega \qquad \frac{1}{2}\left(\frac{2}{5}+1\right)mr^2\left(\frac{v}{r}\right)^2 = mg(R-r)\sin\theta$

$v = \sqrt{\frac{10}{7}g(R-r)\sin\theta} \qquad m\frac{v^2}{(R-r)} = N - mg\sin\theta - qvB$

In limiting condition $N = 0; \theta = 90^\circ$

$qvB = mg + \frac{10}{7}mg = \frac{17mg}{7} \qquad B_{limiting} = \frac{17}{7}\frac{mg}{qv} = 2T$

27. Net flux passing through complete loop is a^2B

Emf induced in loop is $= a^2\left(\frac{dB}{dt}\right) = 20a^2$

Current in loop $i = \frac{20a^2}{12a\lambda}$; λ is resistance per unit length. $i = 2amp$

28. $kr^4(2\pi r) = \mu_0 \int cr^n \times 2\pi r dr,\ 2\pi kr^5 = \left(\frac{\mu_0 C \times 2\pi}{r+2}\right)r^{n+2},\ n+2=5,\ n=3$

29. $\lambda_{th} = \frac{1240}{2} = 620,\ i = \frac{dN}{dt}e = \frac{e\times A}{hc}\int \lambda dI = \frac{eAk\int_{400}^{620}\lambda\ d\lambda}{hc}\ ,\ \frac{1\times 1\times\left(620^{2-}400^2\right)\times k}{1240\times 2}$

$= \frac{220\times 1020}{1240\times 2}\times\frac{0.031}{100} = 2.8\ A \approx 3\ A$

30. For lens, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} - \frac{1}{-4f} = \frac{1}{f} \Rightarrow v = \frac{4f}{3}$.

Lens forms real image of given point source at a distance $\frac{4f}{3}$ from it. If this is at the center of curvature of concave mirror, the rays fall on the mirror normally and retrace their path. Now the distance of mirror $d_1 = \frac{4f}{3} + 2f = \frac{10f}{3}$.

Similarly, if the real image due to lens is formed at the pole of mirror, rays are reflected symmetrically and image is formed at source it self. Now distance of mirror

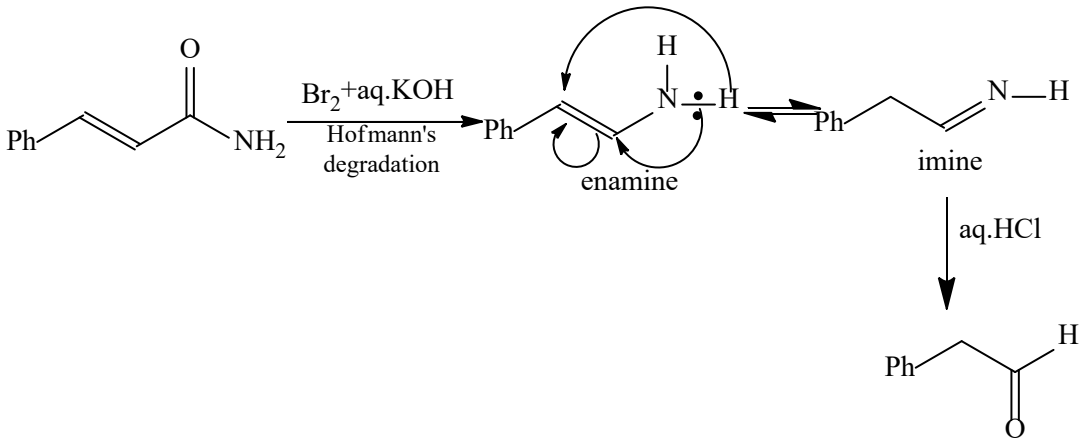
$$d_2 = \frac{4f}{3}.$$

$\frac{d_1}{d_2} = \frac{10}{4} = \frac{5}{2} \Rightarrow n = 5$

31. Conceptual
32. Conceptual
33.
34.

CHEMISTRY

35. Ans: B



Sol:

36. Ans: B

Sol: For azeotropic mixture total vapour pressure is maximum ($+ve$ deviation) or minimum ($-ve$ deviation).

$$P_T = P_A^\circ + 3\left(P_B^\circ - P_A^\circ\right)\chi_B + 2\left(P_A^\circ - P_B^\circ\right)\chi_B{}^2$$

For azeotropic mix $\dfrac{dP_T}{d\chi B} = 0 \Rightarrow \chi_B = \dfrac{3}{4}, \chi_A = \dfrac{1}{4}$

37. Ans: C

Sol: η – factor K_2SO_4 is 2, but when it present as alum η – factor of alum is 8. Thus we have to solve it with molar conductivity.

$$\lambda_m^\circ k = X; \lambda_m^\circ Al^{3+} = 3Y, \lambda_m^\circ SO_4^{2-} = 2Z\ S.cm^2mol^{-1}$$

$$\lambda_m^\circ\left(K_2SO_4.Al_2\left(SO_4\right)_3.24H_2O\right) = 2\lambda_m^\circ K^+ + 2\lambda_m^\circ Al^{3+} + 4\lambda_m^\circ SO_4^{2-}$$

$$= \left(2\times 64 + 82\right) S.cm^2mol^{-1}$$

$$\lambda_{eq}^\circ\left(K_2SO_4.Al_2\left(SO_4\right)_3.24H_2O\right) = \frac{\lambda_m^\circ\left(K_2SO_4.Al_2\left(SO_4\right)_3.24H_2O\right)}{8}$$

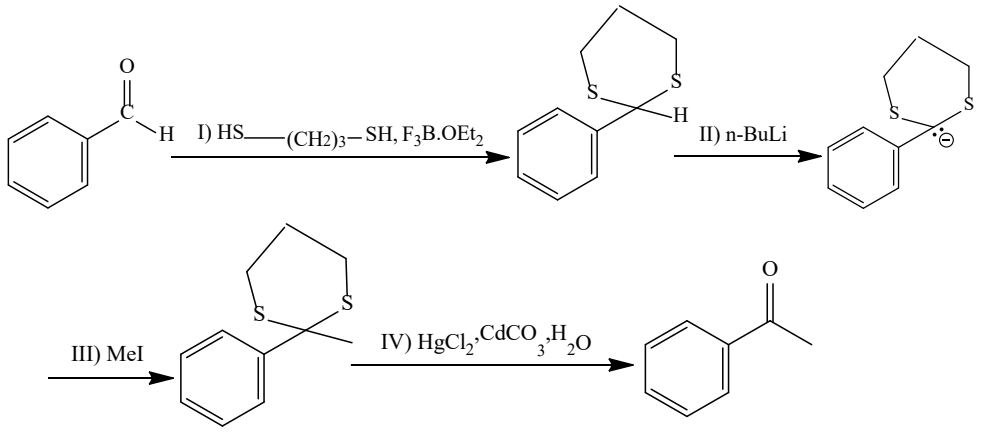
$$= \left[\frac{X}{4} + \frac{3Y}{4} + Z\right] S.cm^2mol^{-1}$$

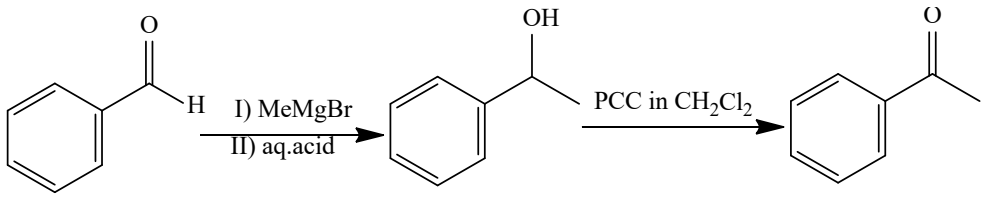
38.

39. Ans: BD

Sol:

B)

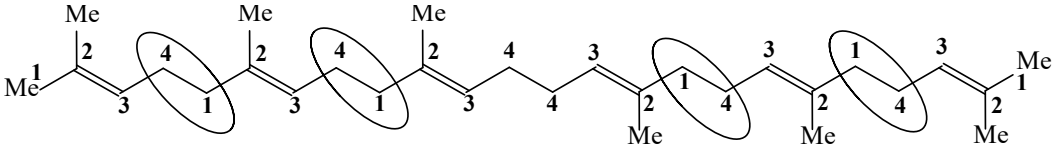


D) 

40. Ans: ABC

41. The problem concerns the standard reduction potentials of $Mn^{3+}(aq), Fe^{3+}(aq) and\ Co^{3+}(aq)$ to the corresponding 2 – 1 ions. All six of the ions are octahedrally coordinated by H_2O molecules in acidic solution. The metals in the 3+ ions are $d^4, d^5 and\ d^6$ species. The metals in the 2+ ions are $d^5, d^6 and\ d^7$ species respectively. These aqua complex are all high spin complexes except $\left[Co(H_2O)_6\right]^{3+}$. The reduction potential of $Fe^{3+}(aq)$ is less than the reduction potential of its neighbours ion the periodic table, which means that, it is harder to reduce to the +2 state than the neighbours. The $Fe^{3+}(aq)$ ion has the high – spin $t_{2g}^3 e_g^2$ configuration. This is respectively stable configuration relative to configurations with 4 or 6 d electrons (note that both the t2g and eg levels of Fe^{3+} are half filled). As a consequences, the $Fe^{3+}(aq)$ ion resist reduction. Reduction of Mn3+ having $t_{2g}^3 e_g^1$ configuration also changes to stable $t_{2g}^3 e_g^2$ config in Mn^{2+}.So Mn^{3+} can be reduced easily to Mn^{2+} but further reduction will be resisted. Co^{2+} is stable in aqueous solution. It will have $t_{2g}^4 e_g^2$ electronic configuration in aqueous solution. Reduction of $Co^{3+}\ to\ Co^{2+}$ changes to $t_{2g}^5 e_g^2$ in which electron is added to t_{2g} orbital that increases CFSE. So can be reduced easily.

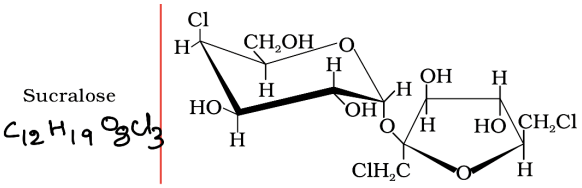
42. Ans: 4

Sol: 

1,4 – addilion is head to tail addition

43. Ans: 397

Sol: NCERT P.No.457



44. Ans: 6

Sol: $\ln\frac{P_1}{P_2} = \frac{\Delta H_{ads}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

Or, $\ln\frac{1.6}{32} = \frac{\Delta H_{ads}}{R}\left(\frac{1}{200} - \frac{1}{250}\right) \Rightarrow \Delta H_{ads} = 600 cal/mol$

45. Ans: 71

Sol: $Co^{3+} + e^- \longrightarrow Co^{2+}; E^\circ = +1.82V$

$\left[Co(CN)_6\right]^{4-} \longrightarrow \left[Co(CN)_6\right]^{3-} + e^-; E^\circ = +0.838V$

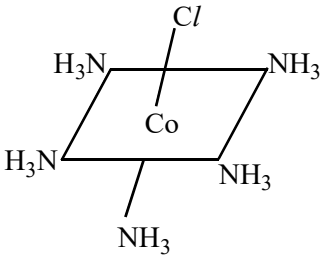
$$\left[Co(CN)_6\right]^{4+} + Co^{3+}2+ \longrightarrow \left[2+Co(CN)_6\right]^{3-} + Co^{2+}; E^{\circ}_{cell} = +2.658V$$

$$E^{\circ}_{cell} = \frac{0.06}{\eta}\log k_c = \frac{0.06}{1}\log\left(\frac{k_f \text{ of}\left[Co(CN)_6\right]^{3+}}{k_f \text{ of}\left[Co(CN)_6\right]^{4-}}\right)$$

$$2.658 = 0.06\log\left(\frac{k_f}{10^{+19}}\right) \Rightarrow 44.3 = \log\left(\frac{k_f}{10^{+19}}\right) \quad \Rightarrow \frac{k_f}{10^{19}} = 2\times10^{44} \Rightarrow k_f = 2\times10^{63}$$

46. By combining with Cl_2, CO gives $CoCl_2, SO_2$ gives SO_2Cl_2, NO gives $NOCl, NO_2$ gives NO_2Cl, PCl_3 gives $PCl_5, FeCl_2$ gives $FeCl_3$ and $CuCl$ gives $CuCl_2$

47.



In this compound to get trans isomer the Cl^- has only one possible position to attack where as to get the cis isomer four positions are available to attack. So the possibility of the formation of cis isomer is about 4 times greater than the formation of trans isomer.

48. Ans: B

Sol: P) $1-$alkyl halide, with poor nucleophile and weak base like H_2O, ROH no reaction

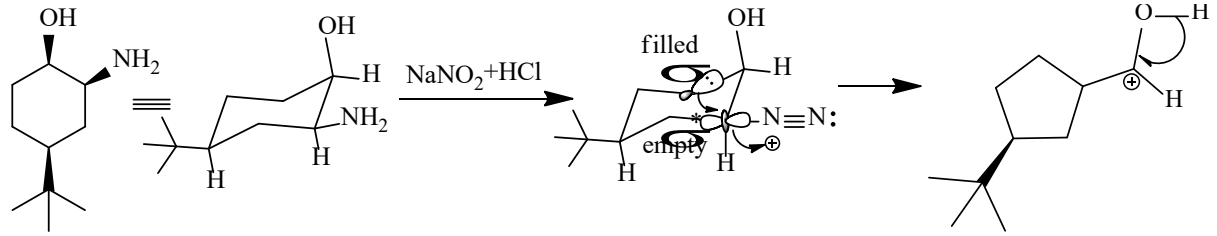
Q) Strong base favour elimination (E_2), with increasing branching at $\alpha-$Carbon and $p-$Carbon possibility for elimination increases.

R) $3^{\circ}-$alkyl halides with weak base poor nucleophiles $(H_2O, ROH....)$ and with weak base strong nucleophile $\left(^{-}CN, I^-, RS^-\right)$ undergo SN^1 at low $T, E,$ at high T.

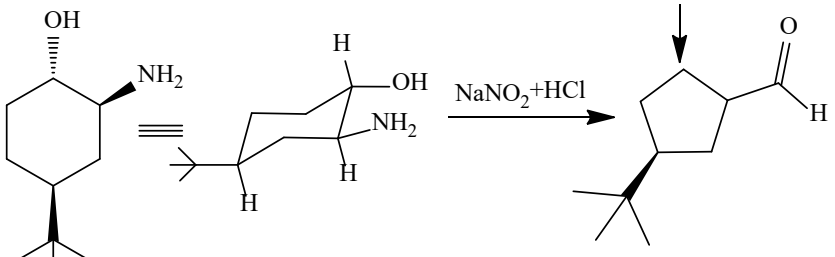
S) Base catalyzed dehydration of aldots $(\beta-\text{hydro Carbonyl Compounds})$ is E_{1CB}.

49. Ans: A

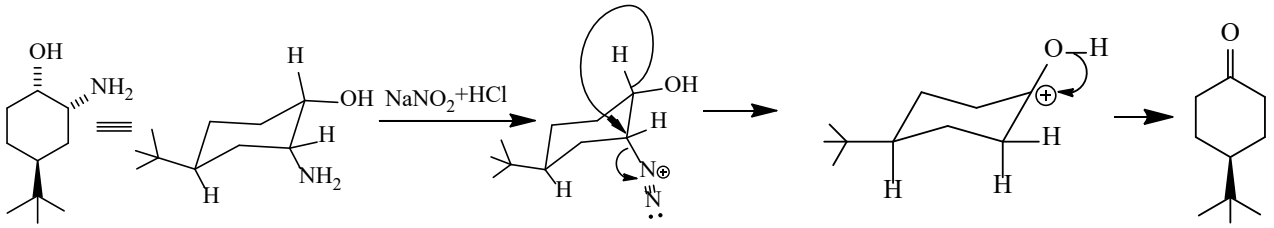
Sol: P)



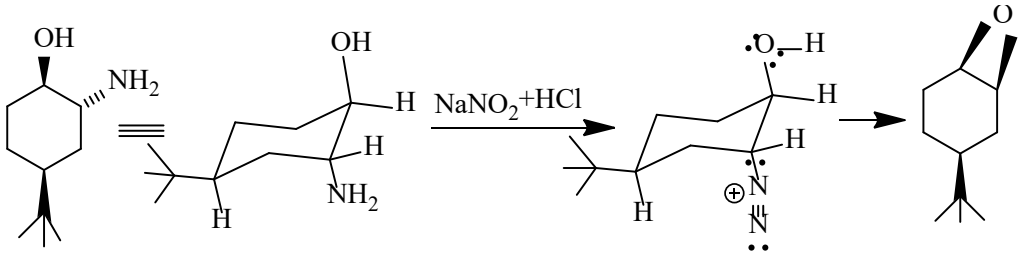
Q)



R)



S)



50. Ans: D
51. Conceptual